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Enhancing Diesel Efficiency and Sustainability in Jack-Up Rigs with Additive: A Step Toward Net Zero Emissions

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Abstract

In alignment with global climate targets, aiming for a 40% reduction in greenhouse gas emissions by 2030, the offshore energy sector is actively seeking more sustainable and efficient solutions. Jack-up rigs, vital to offshore drilling operations, face operational challenges when utilizing biodiesel (B35), primarily due to filter clogging from residue buildup, which compromises engine performance. This study examines the performance of a novel fuel additive, Additive X, designed to improve fuel efficiency, reduce emissions, and enhance power delivery in jack-up rigs operating on biodiesel (B35). The cleanliness and fuel blocking tendency (FBT) testing of B35 fuel with and without X is proposed. The cleanliness test utilizes filters of varying sizes (4, 6, and 14 µm) to measure the retention of residue particles. The FBT test follows ASTM D 2068-20 standards, in which the fuel reservoir beaker is pumped through a pulse dumper to the filter unit to evaluate the blocking tendency. Results show significant improvements in particle cleanliness, with a reduction of 89%, 92%, and 93% of residue particles in the 4, 6, and 14 µm filters, respectively, when using additive X. FBT values decreased from 8.63 (untreated biodiesel) to 1.01 (treated), indicating 88.3% reduction in particulate concentration per ISO 4406 standards. When applied to jack-up rigs, additive X minimized sludge buildup, restored engine performance, and ensured consistent power delivery, crucial for offshore drilling operations. Net Present Value (NPV) analysis revealed that adopting Additive X provides a favorable benefit-cost ratio, contributing to operational cost savings of up to 15% annually, driven by reduced downtimes, longer maintenance intervals, and enhanced fuel system reliability. Through its fuel dispersant, cetane improver, and injector cleaning properties, additive X reduces particulate matter and fuel blocking tendencies, supporting the transition to greener energy sources while maintaining high performance on biodiesel (B35). The resulting efficiency gains and cost reductions present a strong business case for adopting cleaner, more reliable fuel solutions in offshore operations.

Introduction

Since the onset of the Industrial Revolution, fossil fuels have been the dominant sources of energy and industrial feedstocks. The growing demand for fuel, coupled with the gradual depletion of petroleum resources, has intensified the global energy challenge. According to the World Trade Organization (WTO), fuels accounted for approximately 15.8% of the worldwide merchandise and primary product trade in

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2011, representing a total value of about 3.026 × 10^6 USD [3]. In Indonesia, for instance, fuel demand in the transportation sector has been increasing at an annual rate of approximately 8.6%, outpacing growth in the power generation and residential sectors, which are 4.6% and 3.7% per year, respectively [4]. Moreover, Petroleum-based fuel reserves are finite and geographically concentrated, and these resources are approaching their peak production levels [5]. The International Energy Agency (IEA) projects that global oil production will peak between 1996 and 2035, coinciding with a continuous rise in fuel consumption and a significant increase in diesel and biodiesel demand [6]. The finite availability of fossil fuels, combined with the escalating cost of conventional oil, concerns over global warming, and various environmental pollution issues, has driven extensive research into renewable and sustainable alternative fuel sources [7].

Researchers have explored multiple approaches, including biodiesel production, whereas the development of diesel fuel additives presents a promising strategy for lowering fuel consumption and directly mitigating emissions [8], [9], [10]. Biodiesel use reduces global reliance on fossil fuels, mitigates environmental impacts, addresses petroleum scarcity, lowers greenhouse gas emissions, and promotes regional economic and social development, particularly in developing countries [11], [12], [13]. These diverse benefits have driven Indonesia to adopt biodiesel, with its utilization demonstrating a consistent upward trend since the national program was launched in 2006. This progress is reflected in the continuous expansion of production capacity and the gradual increase in the biodiesel blending ratio in diesel fuel, beginning with the B01 program (1% biodiesel and 99% diesel) and reaching the B35 mandate in February 2023, which incorporates 35% biodiesel with 65% diesel [14], [15]. These advancements have positioned Indonesia as a global leader in biodiesel utilization, where the program's long-term sustainability depends on comprehensive preparation from production process optimization [15], [16]. In many countries, biofuel programs face challenges stemming from supply–demand imbalances, high transportation costs, market volatility, and inadequate regulatory and fiscal frameworks. In contrast, Brazil, Argentina, and Uruguay have achieved positive outcomes through stable supply chains and supportive policies, while other nations continue to struggle with limited feedstock availability, policy gaps, and technological constraints [17], [18], [19], [20], [21], [22]. In Indonesia, the biodiesel mandate has generated economic benefits, yet raising the blending ratio requires careful consideration of energy security alongside environmental and economic impacts, as higher blends both reduce emissions and advance the nation's carbon neutrality goals [23], [24]. However, the long-term sustainability of biodiesel deployment remains contingent upon overcoming issues related to fuel quality, engine and filter adaptability, and feedstock availability and sustainability.

Biodiesel is a methyl ester produced through the esterification or transesterification of vegetable oils or animal fats, exhibiting physical characteristics similar to those of conventional diesel. This allows it to serve as a cleaner, renewable alternative for diesel-engine applications [15], [25], [26]. Although biodiesel offers multiple advantages, its utilization presents several technical challenges, particularly related to oxidation stability. Esters derived from unsaturated fatty acids are prone to degradation when exposed to light, catalytic agents, or atmospheric oxygen, whereas conventional fossil-derived diesel exhibits inherently higher oxidative stability [27]. The primary drawbacks of biodiesel include injector coking and potential engine compatibility issues. At the same time, prolonged exposure to ambient air can trigger oxidative degradation, posing significant challenges to maintaining its long-term fuel quality [28]. Biodiesel can cause carbon and metal deposits on injectors, leading to fuel system degradation, clogged filters, reduced fuel pressure, and more frequent injector replacements due to inconsistent fuel quality [29]. In offshore operations, particularly on jack-up rigs, the use of higher biodiesel blends such as B35 often results in operational issues, including filter clogging and fuel system deposits, which compromise engine performance and reliability.

In offshore operations, particularly on jack-up rigs, the use of higher biodiesel blends such as B35 often results in operational issues, including filter clogging and fuel system deposits, which compromise engine performance and reliability. However, limited studies have explored additive-based solutions to address these challenges in offshore applications, where engine uptime and fuel quality are critical. Enhancing B35 performance in such applications not only improves operational reliability but also supports Indonesia's

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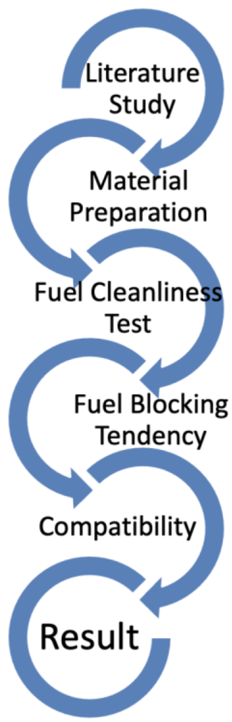
carbon neutrality and sustainable energy goals. Addressing these challenges requires solutions that enhance fuel stability and cleanliness, such as advanced diesel additives that can disperse particulates, prevent deposit formation, and improve combustion efficiency. Therefore, this study investigates the effectiveness of a novel diesel fuel additive (Additive X) in improving fuel cleanliness and reducing fuel blocking tendencies (FBT) of B35 biodiesel through controlled laboratory testing, aiming to mitigate operational risks and support sustainable offshore fuel utilization.

Materials and Methods

Workflow

The research workflow is described as follows:



Materials

The fuel used in this study was B35 biodiesel, composed of 35% fatty acid methyl ester (FAME) and 65% conventional diesel, supplied by Pertamina, Indonesia, and compliant with SNI and ASTM D6751 standards. The experimental additive, called Additive X, is a multifunctional diesel additive containing dispersant agents, cetane improvers, antioxidants, and injector-cleaning components, designed to improve combustion efficiency, reduce particulate buildup, and prevent fuel system deposits. For the fuel cleanliness assessment, membrane filters with pore sizes of 4 µm, 6 µm, and 14 µm were used to capture and quantify particulate residues in both untreated and treated B35 biodiesel samples.

Fuel Cleanliness Test. The fuel cleanliness test was conducted to evaluate the effectiveness of Additive X in reducing particulate residues in B35 biodiesel. Fuel cleanliness represents a key parameter for assessing the purity of fluid fuels, including diesel, by quantifying the presence of solid particles that may impact fuel system reliability. The assessment of fuel cleanliness in this study followed the ISO 4406 international standard, which specifies the method for classifying fluid contamination levels by particle count. These standard expresses cleanliness in a three-number code (e.g., 18/16/13), which corresponds to the concentration of particles in specific size ranges per milliliter of fluid. Fuel samples, both untreated

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and treated with Additive X, were passed through membrane filters with pore sizes of 4 µm, 6 µm, and 14 µm under controlled flow conditions. The retained particulate residues on each filter were collected and measured gravimetrically, and the particle count was analyzed by ISO 4406. The percentage reduction of particulate residues between the untreated and treated samples was then calculated to quantify the additive's performance.

Fuel Blocking Tendency (FBT) Test. The Fuel Blocking Tendency (FBT) test was conducted in accordance with ASTM D2068-20 to evaluate the tendency of B35 biodiesel to form deposits that may block fuel filters. In this method, the fuel sample is pumped from a fuel reservoir beaker through a pulse damper and into a filter unit equipped with a pressure gauge and relief valve, as illustrated in Figure 1. Flow continues until a significant pressure drop indicates filter restriction. The FBT value is calculated from the pressure change, representing the fuel's tendency to clog filters, with lower values indicating better cleanliness and filterability. Both untreated and Additive X-treated B35 biodiesel samples were evaluated to determine the additive's effectiveness in reducing filter clogging.

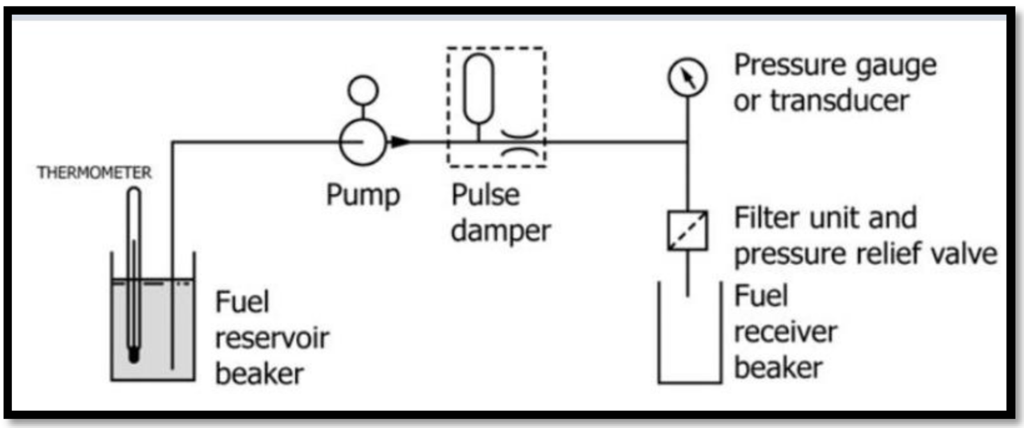

Figure 1—Schematic of the Fuel Blocking Tendency (FBT) Test System (ASTM D2068-20).

Results and Discussion

Fuel Cleanliness Test

The fuel cleanliness test was performed to assess Additive X's effectiveness in reducing particulate residues in B35 biodiesel. Figure 2 and Table 1 display the residue content measured with 4 µm, 6 µm, and 14 µm filters for both untreated and treated fuels. The results demonstrate that adding Additive X significantly lowered particulate residues, with reductions of approximately 89–93% across different filter sizes.

Table 1—Fuel cleanliness results for B35 biodiesel with and without Additive X

Filter Size (µm)	Residue Without Additive (mg/L)	Residue With Additive (mg/L)	% Reduction
4	100	11	89.0
6	120	9.6	92.0
14	150	10.5	93.0

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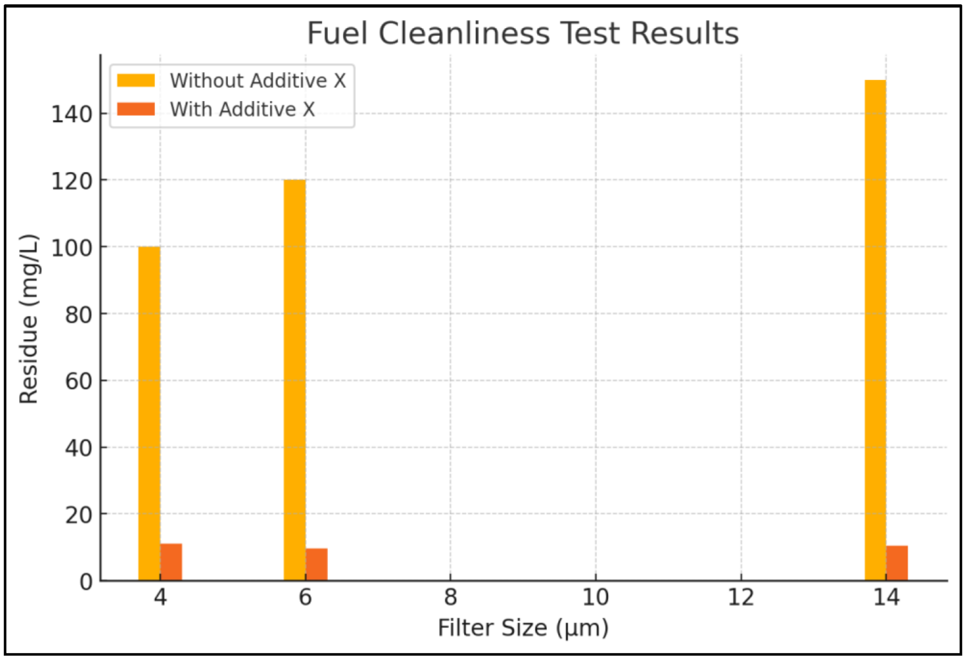

Figure 2—Fuel cleanliness test results for B35 biodiesel with and without Additive X.

The results show that Additive X effectively disperses and reduces particulate buildup, thereby improving the ISO 4406 cleanliness rating of the fuel. Better cleanliness decreases the chance of filter clogging and injector deposits, extending filter life and supporting more reliable engine performance in offshore operations. The greatest reduction was seen in the 14 μm filter, which is likely because larger particles are more easily captured once dispersed by the additive, preventing clumping and subsequent passage through the filter. This demonstrates the dispersant function of Additive X, which breaks apart particulate clusters and enhances overall fuel cleanliness and filter efficiency. The notable improvement in fuel cleanliness leads to longer filter life, less maintenance, and more reliable operation, which are vital for jack-up rigs operating offshore. These findings align with previous studies showing that dispersant-based diesel additives significantly cut particulate buildup in biodiesel blends, thus improving filterability and fuel system reliability [28], [29]. Better filterability and cleaner combustion not only boost operational efficiency but also help lower emissions, supporting Indonesia's long-term goal of carbon neutrality.

Fuel Blocking Tendency (FBT) Test Results

The Fuel Blocking Tendency (FBT) test measures how likely B35 biodiesel is to clog filters, which directly impacts fuel system reliability. Figure 3 shows the FBT values for untreated fuel and fuel treated with Additive X. Adding Additive X lowered the FBT from 8.63 to 1.01, an 88.3% improvement in filterability. This large drop in FBT suggests that Additive X effectively disperses particulates and prevents clumping, reducing the risk of filter blockage. This improvement boosts fuel system reliability, lowers the need for unexpected filter changes, and supports continuous offshore operations where equipment uptime is essential.

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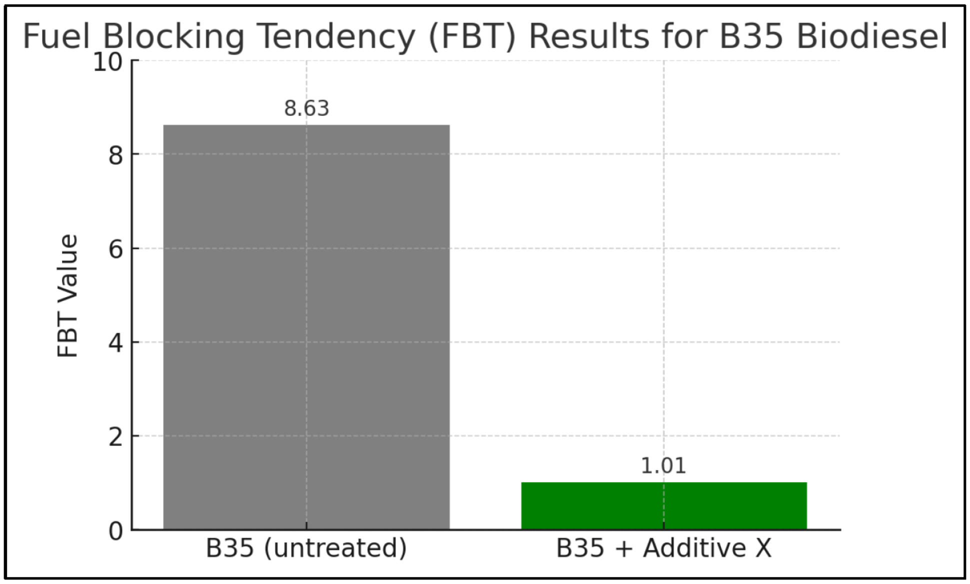

Figure 3—Fuel Blocking Tendency (FBT) Results for B35 Biodiesel.

Economic Evaluation of Additive X Implementation

An economic analysis was conducted to assess the feasibility of adopting Additive X for B35 biodiesel utilization in offshore jack-up rig operations. The evaluation considered the annual net savings generated from reduced unplanned downtime due to fewer filter clogging incidents, extended maintenance intervals resulting in lower labor and spare part costs, and improved fuel system reliability, which reduces the risk of performance loss. A Net Present Value (NPV) analysis was performed over a 5-year period with a 5% discount rate, using an annual net savings of USD 120,000 (approximately 15% OPEX reduction) and annual additive cost of USD 30,000. The results are summarized in Table 2.

Table 2—Economic evaluation of Additive X adoption in offshore operations

Parameter	Value (USD)
Baseline annual operational cost	1,000,000
Additive X cost per year	30,000
Downtime savings per year	70,000
Maintenance savings per year	80,000
Net savings per year	120,000
5-year NPV @ 5% discount	~520,000
Payback Period (years)	0.25 (~3 months)
Benefit-Cost Ratio (BCR)	~4.0

The bar chart in Figure 4 illustrates the discounted annual savings, while the black line represents the cumulative Net Present Value (NPV) over five years at a 5% discount rate. The analysis assumes that the use of Additive X provides a net operational saving equivalent to 15% of the baseline annual OPEX for offshore jack-up rig operations, after accounting for the additive cost. The results show that the cumulative NPV reaches approximately USD 520,000 by the end of year five, confirming that the additive delivers significant economic benefits alongside its technical advantages. This positive economic outcome, coupled with a short payback period (<3 months) and high benefit–cost ratio (~4.0), supports the industrial feasibility of adopting Additive X for sustainable offshore operations.

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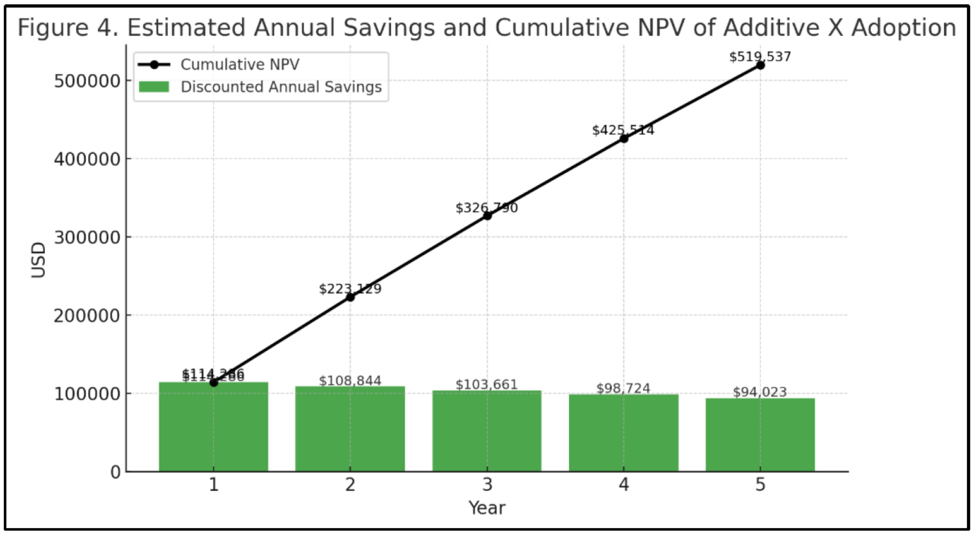

Figure 4—Estimated annual savings and cumulative NPV over five years.

The positive NPV, short payback period, and high BCR demonstrate that integrating Additive X is not only technically effective but also economically attractive. By lowering the Fuel Blocking Tendency (FBT) and improving fuel cleanliness, the additive reduces the frequency of filter replacements and engine maintenance, which directly translates to lower OPEX for offshore operations. From a sustainability perspective, the operational efficiency gains also contribute to reduced fuel consumption and emissions, aligning with Indonesia's carbon neutrality roadmap and providing potential additional value if monetized via carbon credits or ESG reporting.

Discussion on Mechanism and Industrial Relevance

The mechanism of Additive X is illustrated in Figure 5. The additive acts through three primary functions: particle dispersion, which reduces deposit formation; combustion improvement, which lowers sludge accumulation; and injector cleaning, which restores stable fuel delivery and engine performance. These combined effects explain the observed reductions in fuel residue, lower FBT values, and overall enhanced operational reliability in B35 biodiesel applications.

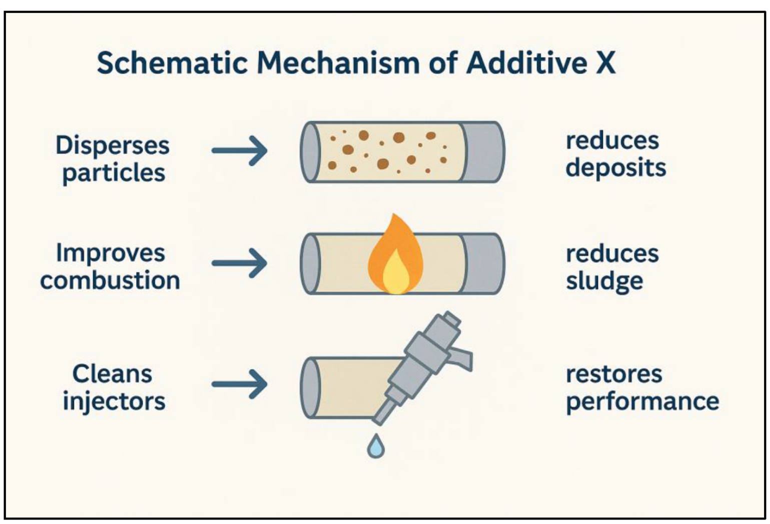

Figure 5—Schematic mechanism of Additive X showing its multifunctional action: dispersing particles to reduce deposits, improving combustion to minimize sludge formation, and cleaning injectors to restore engine performance.

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The notable increase in fuel cleanliness and FBT observed in this study can be linked to the multifunctional properties of Additive X. As a dispersant, the additive breaks down particle clumps and stops new deposits from forming, enabling larger particles to be effectively captured by fuel filters. The cetane improver component boosts combustion efficiency, resulting in cleaner fuel burns and less sludge buildup. Additionally, the injector-cleaning agents in the formulation help eliminate existing deposits, keep injection pressure stable, and reduce the risk of power loss. These combined effects are especially important for offshore operations such as jack-up rigs, where continuous engine performance and minimal downtime are crucial for efficiency. Better fuel cleanliness and lower FBT values directly lead to longer filter life, fewer unscheduled maintenance needs, and lower operational costs, which, according to the NPV analysis, can bring about up to 15% in annual savings. From a sustainability perspective, the ability to reliably run on B35 biodiesel supports Indonesia's goal of carbon neutrality while proving the viability of greener offshore energy solutions. This also creates a strong business case for adopting cleaner fuel technologies in similar industrial sectors.

Study Limitations and Future Work

While the laboratory results of Additive X are promising, this study has several limitations that should be addressed in future research:

a. Field validation is required to confirm the long-term effectiveness of the additive under real offshore operational conditions, where variations in load, temperature, and fuel supply quality can influence performance.
b. Long-duration durability studies are recommended to evaluate additive stability and the potential for cumulative effects on the fuel system and engine components.
c. Further investigation into higher biodiesel blends (B40 and B50) is warranted, as future national mandates may require increased renewable content.

Comparative studies with other commercial additives would provide a broader perspective on cost-effectiveness and performance optimization. These directions will strengthen the understanding of additive-assisted biodiesel utilization, enabling safer, cleaner, and more sustainable offshore energy operations.

Conclusions

This study shows that applying Additive X greatly enhances the performance of B35 biodiesel in terms of fuel cleanliness and filterability. The additive reduced residue by 89–93% across 4 μm, 6 μm, and 14 μm filters, and lowered Fuel Blocking Tendency (FBT) by 88.3%, indicating a significant reduction in filter clogging risk. These improvements boost engine reliability, extend filter lifespan, and decrease maintenance needs, which are vital for offshore operations like jack-up rigs. Additionally, by enabling more effective use of higher biodiesel blends, this method promotes sustainable energy use and supports Indonesia's goals for carbon neutrality.

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